

SAP ID: 590020087

Data Analysis and Visualization

Question: Create numpy array to find sum of all elements in an array?

Code:

```
Array: [ 7 12  9  4 11 25 15]
Sum of all elements: 83
```

Question: Create numpy array of (3, 3) dimension. Now find sum of all rows and columns individually. Also find 2nd maximum element in the array.

Code:

```
3x3 Matrix [[ 9 12 15]
 [ 7 12  9]
 [ 4 11 25]]
Sum of Rows: [36 28 40]
Sum of Columns: [20 35 49]
Second Maximum element: 15
```

Question: Perform Matrix multiplication of any 2 $n \times n$ matrices.

Code:

```
Matrix A: [[ 7 12  9]
 [11 20 25]
 [ 3  9 12]]
Matrix B: [[ 7 12  9]
 [11 20 25]
 [ 3  9 12]]
MatrixA x MatrixB [[208 405 471]
 [372 757 899]
 [156 324 396]]
```

Question: Write a Pandas program to get the powers of an array values element-wise.

Code:

```
Base Values: 0 0
1 1
2 2
3 3
4 4
5 5
6 6
Name: base, dtype: int64
Power Values: 0 1
1 1
2 4
3 27
4 256
5 3125
6 46656
dtype: int64
Powers of an array values element-wise: 0 1
1 1
2 4
3 27
4 256
5 3125
6 46656
dtype: int64
```

Question: Write a pandas program to get the first 3 rows of a given Dataframe.

Code:

Complete DataFrame:				
	name	score	attempts	qualify
a	Anastasia	12.5	1	yes
b	Dima	9.0	3	no
c	Katherine	16.5	2	yes
d	James	NaN	3	no
e	Emily	9.0	2	no
f	Michael	20.0	3	yes
g	Matthew	14.5	1	yes
h	Laura	NaN	1	no
i	Kevin	8.0	2	no
j	Jonas	19.0	1	yes

First three rows:

	name	score	attempts	qualify
a	Anastasia	12.5	1	yes
b	Dima	9.0	3	no
c	Katherine	16.5	2	yes